

discriminating characteristics either. However, the feedback loop from which Tay was supposed to learn had a flaw.

Racial discrimination in Amazon's same-day delivery

In early 2016, Amazon rolled out same-day delivery to its Prime programme subscribers, but only for a select group of American cities and for a select group of neighbourhoods. These neighbourhoods were the ones where concentration of Prime subscribers was large enough to justify operational costs.

However, it soon became an issue when people realised that predominantly non-white neighbourhoods were primarily excluded from this offering. Customers from these areas were not happy.

When this issue was highlighted in several forums and the media, Craig Berman, Amazon's vice president for global communications, asserted, "When it comes to same-day delivery, our goal is to serve as many people as we can." He continued, "Demographics play no role in it. Zero." From Amazon's standpoint, this seemed to be a logical approach from the cost and efficiency perspective, such that they would prioritise areas with most existing paying members over the others.

A solely data-driven approach that looked at numbers instead of people did reinforce long-entrenched inequality in access to their services. For people who were (inadvertently) excluded, it did not make much difference. The fact that racial demographics tend to correlate with location or zipcode did result in indirect discrimination.

Apparently, in scenarios where zipcodes are considered as one of the data points for any decision-making,



National Transportation Safety Board investigators examine Uber's self-driving vehicle that was involved in a fatal accident in Tempe, Arizona, in March 2018 (Credit: <https://cnet4.cbsistatic.com>)

there will always be some inherent bias. It is because zipcodes usually represent tightly-knit communities of one kind. Perhaps better sampling of input data could help to a certain extent in minimising this risk, but it would be far from being absolute zero.

I remember about ten years ago, Indian banks used this (zipcode-based) strategy to exclude applicants from credit card or personal loan applications. I do not know if they are continuing these practices still.

Uber's autonomous car kills a pedestrian

An Arizona pedestrian was killed by an automated vehicle owned by Uber, in 2018. A preliminary report released by National Transportation Safety Board (NTSB) in response to the incident stated that there was a human presence in the automated vehicle, but the human was not in control of the car when the collision occurred.

An accident usually implies several things going wrong at once and, hence, it is difficult to investigate as well as ascertain final accountability and root

cause. Various reasons contributed to this accident, including poor visibility of the pedestrian, inadequate safety systems of the autonomous car and lack of oversight by the human assistant.

The legal matter was eventually settled out of courts, and further details were not released. However, issues around liability present significant complexity due to multiple parties and actors involved. On this occasion, the vehicle was operated by Uber but was under the supervision of a human driver who was an Uber employee, and it was operated autonomously using components and systems designed by various other tech companies.

Responsibility attribution with an AI system poses a significant dilemma and, unfortunately, there are no set standard guidelines to make it more transparent.

On occasions like these, where human lives are directly affected, clarity is undoubtedly needed along with a universal framework to help in a standardised approach to deal with such matters.